Refer attached commented drawing.

DOCUMENT REVIEW DETAILS

1 No Comments. **Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00**

[✓ 2] Proceed with manufacture / fabrication as per the commented document. Revised document required.

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V Void.

Signature with Date **03-SEP-2021**

Name of Reviewer **AJEET KUMAR (B565)**

Page No. **10 of 19**

हिन्दुस्तान पेट्रोलियम HP इंजीनियर्स इंडिया लिमिटेड (भारत सरकार का उपक्रम) ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)	**Residue Upgradation Facility (RUF)** **EPCC-3 Package for** **Visakh Refinery Modernization Project (VRMP)**	L&T Hydrocarbon Engineering **L&T- CHIYODA LIMITED**
Title: ~~Data Sheet With~~ **GAD for Dock Leveller** (504-LZ-930 A/B) **Doc. No.: B016-RUF-LT-504-MP-GN-NTPL(1)-13002**		**Rev. No.: 00** **Page No: 1 of 4**

OWNER	:	**HINDUSTAN PETROLEUM CORPORATION LIMITED (HPCL)**
PMC	:	**ENGINEERS INDIA LIMITED, NEW DELHI (EIL)**
UNIT	:	**504**
PMC JOB NO.	:	**B016**
EQUIPMENT DESCRIPTION	:	**DOCK LEVELLER**
EQUIPMENT TAG	:	504-LZ-930 A/B
P. R. NO.	:	B016-RUF-LT-504-MP-PR-0014
VENDOR NAME	:	**NIHVA TECHNOLOGIES PVT LTD**
VENDOR DOCUMENT NO.	:	**NIHVA_20-21_0157**

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☑ R ☐ I

ISSUED FOR
☐ INFORMATION
☑ REVIEW
☐ CONSTRUCTION
☐ AS BUILT</td>
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PPP/
MRS</td><td>04-Jun-21</td><td></td><td></td><td></td></tr>
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00	Issued for Approval	Vikrant J	Amit K	Amit K			31-05-2021
Revision No.	**Description**	**Prepared by**	**Reviewed by**	**Approved by**	**Reviewed by**	**Approved by**	**Approved Date**

DOCUMENT REVIEW DETAILS

Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00

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Name of Reviewer AJEET KUMAR (B565)

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REVISION HISTORY SHEET

REV. NO.	REV. DATE	REVISED PAGE	REVISION DESCRIPTION
00	31-05	NA	ISSUED FOR APPROVAL

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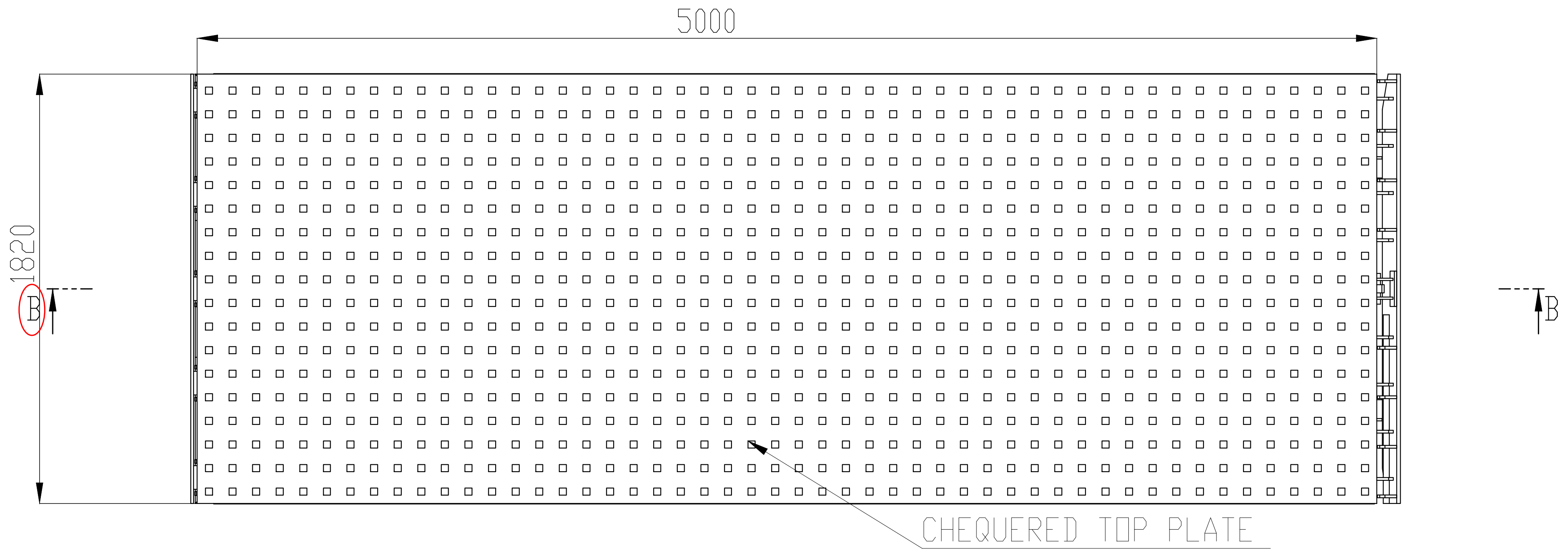
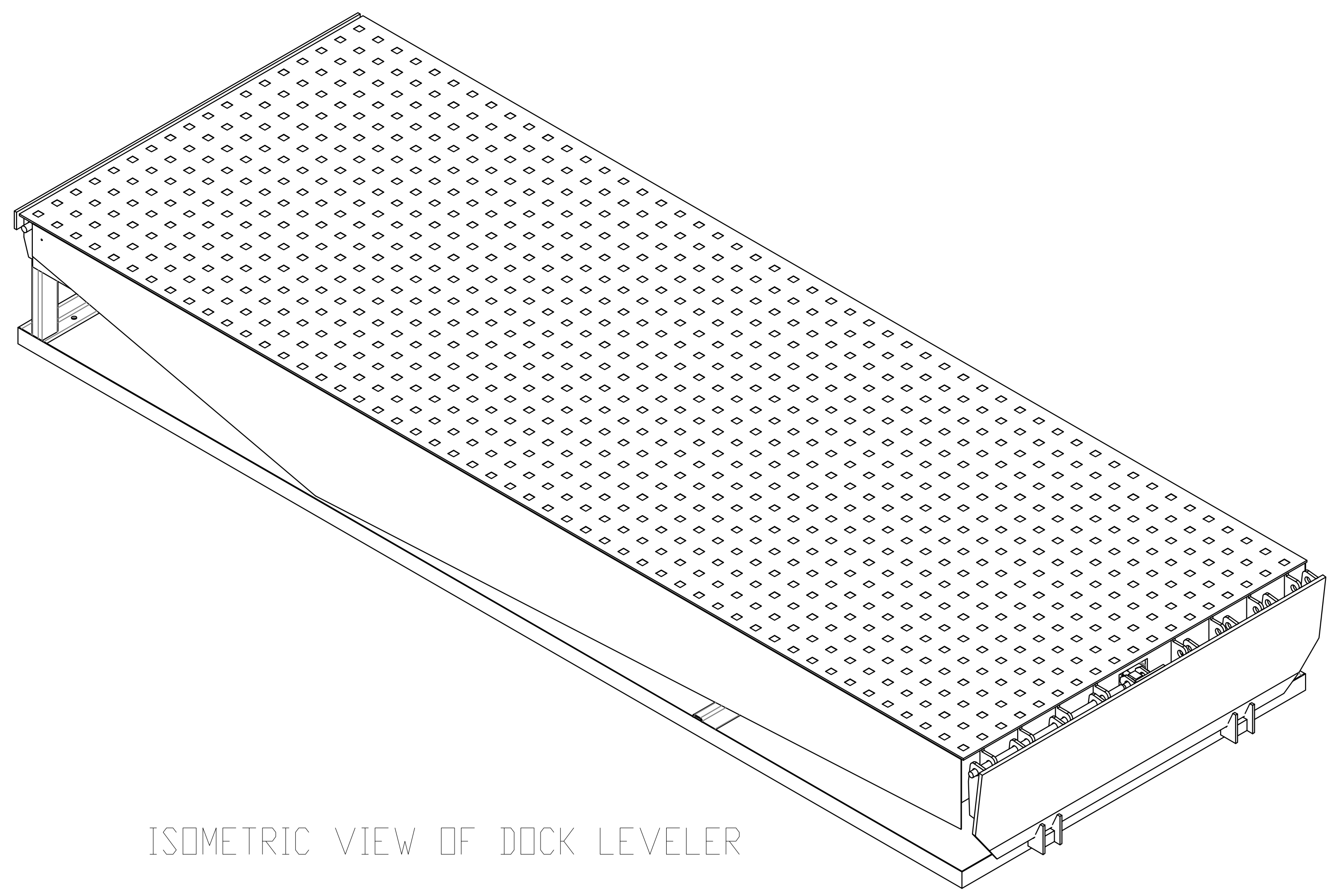
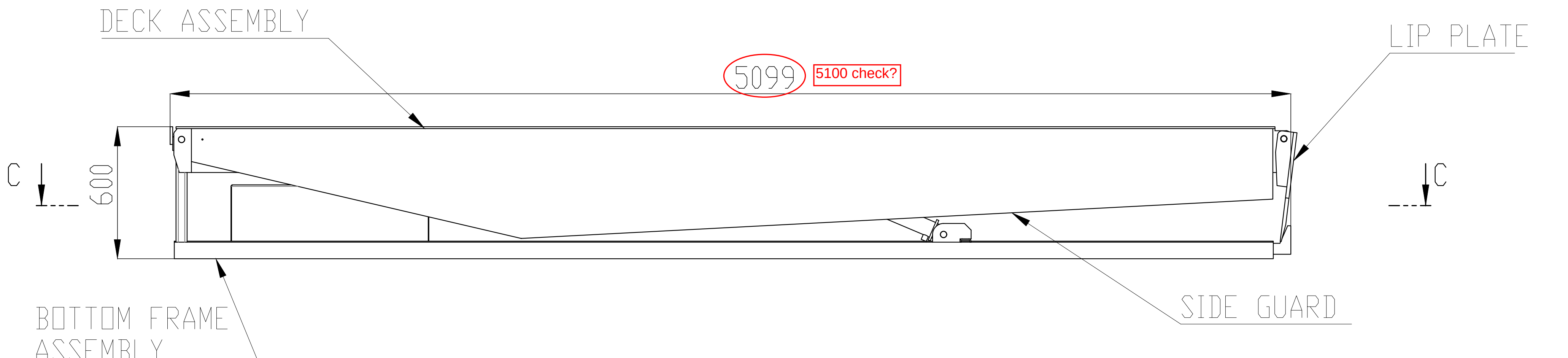
Signature with Date **03-SEP-2021**
Name of Reviewer **AJEET KUMAR (B565)**
Page No. **12 of 19**



DOCK LEVELER TOP VIEW

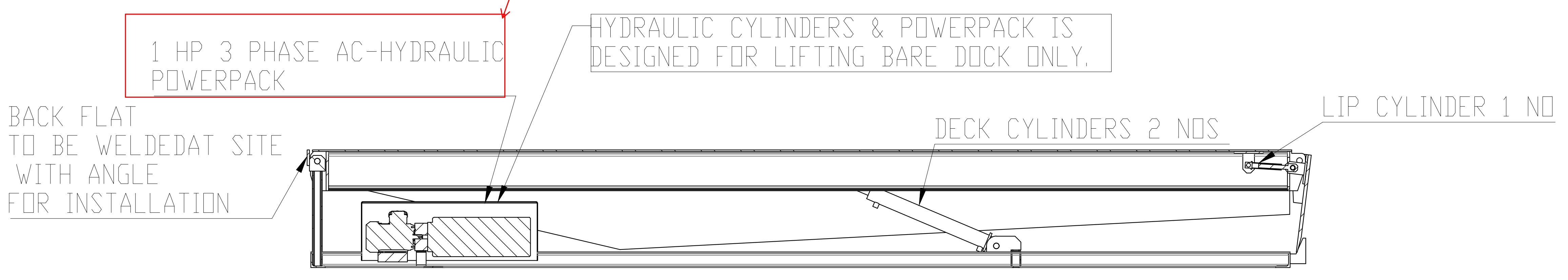


ISOMETRIC VIEW OF DOCK LEVELER

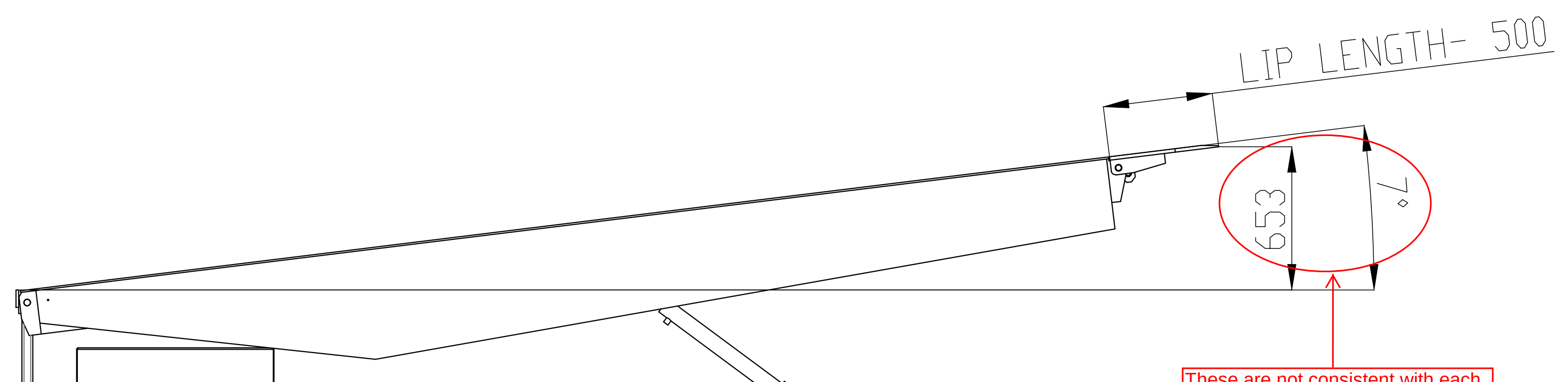


DOCK LEVELER SIDE VIEW

Is hydraulic power pack common for both deck cylinders & lip cylinder? If yes, please indicate clearly.

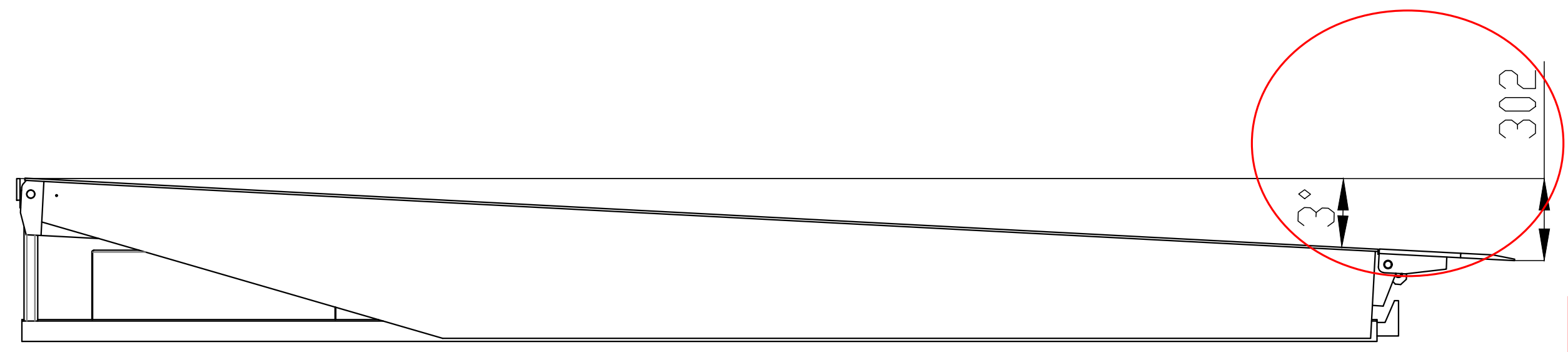

1 HP 3 PHASE AC-HYDRAULIC POWERPACK
HYDRAULIC CYLINDERS & POWERPACK IS DESIGNED FOR LIFTING BARE DOCK ONLY.
BACK FLAT TO BE WELDEDAT SITE WITH ANGLE FOR INSTALLATION
DECK CYLINDERS 2 NOS
LIP CYLINDER 1 NO

SECTION B-B
SCALE 1 : 15



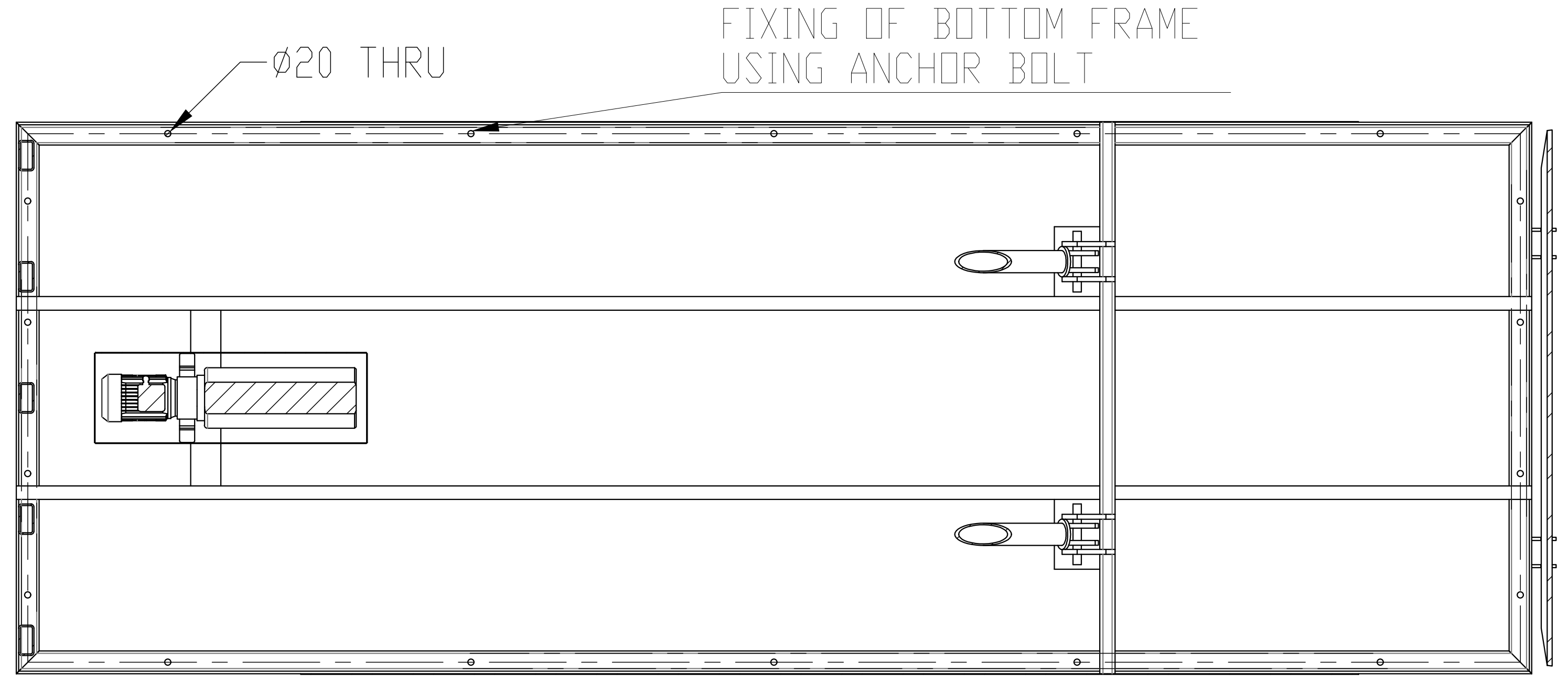
These are not consistent with each other. Please check & update. Maximum slope indicated in attached data sheet shall also be accordingly checked & updated.

DECK UP HEIGHT - 600 MM



These are not consistent with each other. Please check & update. Maximum slope indicated in attached data sheet shall also be accordingly checked & updated.

DECK DOWN HEIGHT - 300 MM

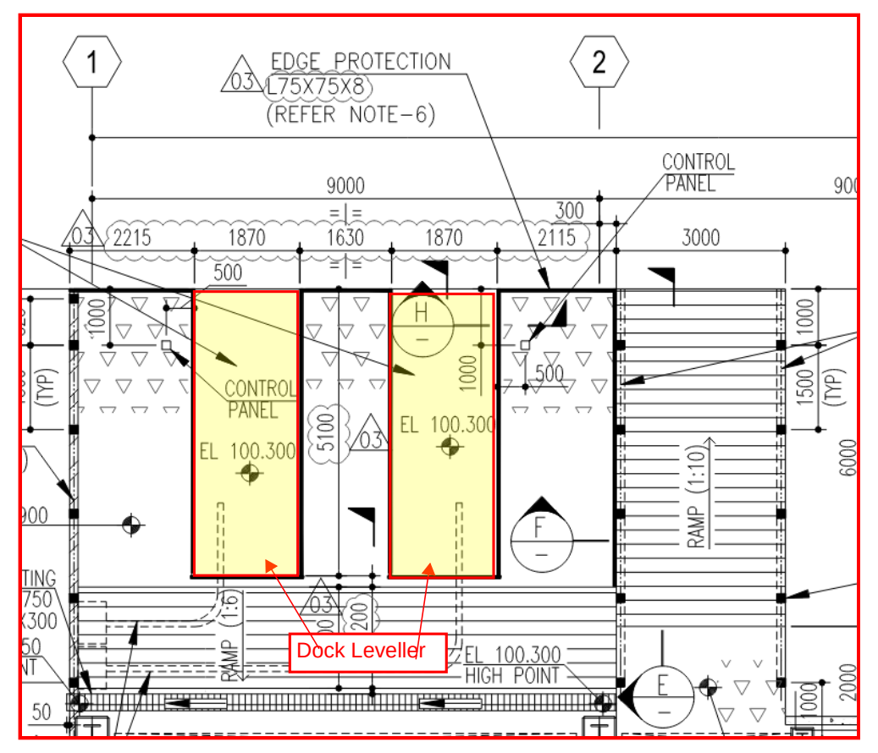


SECTION C-C
SCALE 1 : 15

TECHNICAL SPECIFICATIONS OF PIT MOUNTED ELECRO-HYDRAULIC DOCK LEVELER -

1. SIZE OF DOCK LEVELER = 1820mm(W) X 5000mm~~(W)~~ (L)
2. OVERALL SIZE OF PRODUCT = 1820mm(W)X5100mm(L)X600mm~~(H)~~ (D)
3. CAPACITY = 6 TON POINT LOAD
4. LIP PLATE SIZE = 500mm(L)X1820mm(W)
5. DECK PLATE = CHEQUERED SHEET 1820mm(W)x5000mm(L)
6. DECK & LIP UP=DOWN OPERATION BY HYDRAULIC CYLINDER
7. DECK UP HEIGHT=(+) 600 MM
8. DECK DOWN HEIGHT=(-) 300MM
9. HEIGHT OF DOCK LEVELER FROM ROAD LEVEL= 900 mm ,
10. OPERATION CONTROL BY PUSH BUTTONS ON CONTROL PANEL MOUNTED ON STAND NEAR DOCK LEVELER
11. ELECTRICAL SUPPLY OF EQUIPMENT = 415 VOLT 3 PHASE AC , 50HZ
12. PAINT = FIROZI BLUE
13. MATERIAL OF CONSTRUCTION = MILD STEEL AS PER IS STANDARD
14. MANUFACTURING STANDARD OF DOCK LEVELER =EN1398
15. ELECRICAL MOTOR COMPLIANCE ZONE 1-IIC-T3 HAZARDOUS AREA . FOR OTHER EQUIPMENT=PESO CERTIFICATION
16. PIT SIZE = 1870 mm(W)X5100mm(L)x600 mm(D)
17. DOCK LEVELER FIXING IN PIT BY ANCHOR BOLT .
18. REQUIRED PIT & ALL CIVIL WORK FOR MOUNTING OD DOCK LEVELER IS IN ~~CLIENT~~ LTHE SCOPE .
REFER PIT DRAWING FOR CONSTRUCTION OF PIT

Provide reference document number



EIL Comments:
1. Include a sheet showing both dock levellers together for clarity w.r.t. relative location.
2. Tabulate BOM for one no. dock leveller.
3. "Client" wherever mentioned in the drawing shall be replaced by "LTHE".
4. Other comments as marked.

List here "Civil Drawing for Dock Leveller (504-LZ-930A/B)"

DRAWING NO.	DRAWING TITLE	REV

REFERENCE DOCUMENTS / DRAWINGS

DRG. CATEGORY: (USE TICK MARK) R I
SCH. DATE OF SUBMISSION
DOCUMENT STATUS
STATEMENT OF SUB: 1st 2nd 3rd
☐ FINAL FOR CODE-1 ☑ AS BUILT
ISSUED FOR
☐ AS BUILT ☐ INFORMATION
☐ CONSTRUCTION ☐ REVIEW
SIGN: DATE:

EIL REVIEW/APPROVAL CODE:
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NAME DICIPLINE SIGN. DATE

REV.	DESCRIPTION	DRN	CHECKED	APPROVED	REVIEWED	APPROVED	DATE
00	ISSUED FOR APPROVAL	SMS	VJ	AK			

OWNER: HINDUSTAN PETROLEUM CORPORATION LIMITED
VISAKH REFINERY MODERNIZATION PROJECT
EPCC-03 PACKAGE FOR RESIDUE UP-GRADATION FACILITY (RUF)

PMC : इंजीनियर्स इंडिया लिमिटेड (भारत सरकार का उपक्रम) ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

LSTK CONTRACTOR: L&T Hydrocarbon Engineering

DETAILED ENGINEERING CENTER: L&T - CHIYODA LIMITED

VENDOR : NIHVA TECHNOLOGIES PVT LTD
VENDOR PO NO: REFINARY/75000-82256/VMM
VENDOR DWG. NO: NIHVA_20-21_0157

DATE 31.05.21
SCALE 1:1
TITLE: ~~DATASHEET WITH~~ GAD FOR DOCK LEVELER (504-LZ-930A/B)
DEPT: DESIGN
DRN BY: SMS
CHK BY: VJ
APP BY: AK
PROJECT: L&T-RUF
PWO NO.:
DRG. NO.: B016-RUF-LT-504-MP-GN-NTPL(1)-13002
SH.1 OF 1
REV. 00
CAD REVISE THIS DRG. USING CAD SYSTEM ONLY FILE:

L&T - CHIYODA LIMITED
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COMMENTED BY	REVIEWED BY	APPROVED BY	DATE

THIS REVIEW / APPROVAL DOES NOT RELIEVE THE VENDOR / MANUFACTURER OF THE FULL RESPONSIBILITY AS TO DESIGN CORRECTNESS OF DIMENSIONS & COMPLIANCE WITHALL APPLICABLE REGULATIONS AND WITH THE REQUIREMENTS MADE IN PURCHASER SPECIFICATIONS.

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Signature with Date 03-SEP-2021
Name of Reviewer AJEET KUMAR (B565)
Page No. 13 of 19

DATA SHEET FOR DOCK LEVELLER - (504-LZ-930A/B)

Sr No	Parameter	Remark
1	Required Capacity	06 Ton
2	Leveller operation mode	AC power pack hydraulically operated
3	Lip operation mode	hydraulic
4	Operation control	by push buttons and control panel
5	~~Loading platform~~ height of dock leveller from road Level	900 mm
6	Travel above & below dock level	600 mm & 300 mm
7	Platform Size (W x L)	1820 mm x 5000 mm
8	Lip size (W x L)	1820 mm x 500 mm
9	Maximum slope can be achieved	7 Degree

Both deck up & deck down slopes to be indicated. Also refer comments on GAD in this connection.

* Please refer Attachment-1 for standard catalog of Dock Leveller.

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Signature with Date **03-SEP-2021**

Name of Reviewer **AJEET KUMAR (B565)**

Page No. **14 of 19**